

COMPARISON OF SURFACTANT TYPES AND THEIR APPLICATIONS

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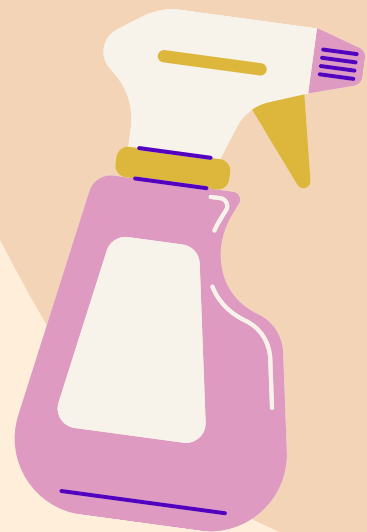
HOW DO ANIONIC, NONIONIC, CATIONIC, AND AMPHOTERIC SURFACTANTS DIFFER?

TYPE	CHARGE ON HEAD GROUP	EXAMPLE	COMMON USES
ANIONIC	NEGATIVELY CHARGE (-)	SODIUM LAURYL SULFATE (SLS)	LAUNDRY DETERGENT, SHAMPOO
CATIONIC	POSITIVELY CHARGED (+)	CETYLTRIMETHYLAMMONIUM BROMIDE (CTAB)	FABRIC SOFTENERS, HAIR CONDITIONER
NONIONIC	NO CHARGE	ALCOHOL ETHOXYLATES	SHAMPOOS, COSMETICS
AMPHOTERIC	BOTH + AND - CHARGES DEPENDS ON PH	COCAMIDOPROPYL BETAMINE	BODY SHAMPOOS, FACIAL CLEANSERS

WHY ARE SOME USED IN LAUNDRY VS. SHAMPOOS VS. DISINFECTANTS?



- **Laundry detergents:** Use anionic surfactants (e.g., sodium lauryl sulfate) for strong cleaning power.
- **Shampoos and body washes:** Use nonionic or amphoteric surfactants (e.g., cocamidopropyl betaine) because they are gentle on skin and hair.
- **Disinfectants:** Use cationic surfactants (e.g., quaternary ammonium compounds) for their antimicrobial properties.



WHICH TYPE IS MOST ENVIRONMENTALLY FRIENDLY?

1

Non-ionic & Amphoteric

- break down easily
- environmentally friendly
- biogradable

2

Anionic

- clean well but take longer time to break down

3

Cationic

- used in disinfectants
- least eco-friendly
- can harm fish & microorganism

THANK
YOU

